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Studierichting: Informatica
Oefeningenreeks 4A nr. 7.5

$$\int \frac{\sqrt{x^3}+\sqrt[3]{3x}}{\sqrt{2x}}dx$$
$$= \frac{1}{\sqrt{2}} \cdot \int \frac{x^{\frac{3}{2}}}{x^{\frac{1}{2}}}dx - \frac{\sqrt[3]{3}}{\sqrt{2}} \cdot \int \frac{x^{\frac{1}{3}}}{x^{\frac{1}{2}}}dx$$
$$= \frac{\sqrt{2}}{2} \cdot \int x dx - \frac{\sqrt{2}\sqrt[3]{3}}{2} \cdot \int x^{-\frac{1}{6}} dx$$
$$= \frac{\sqrt{2}x^2}{4} - \frac{\sqrt{2}\sqrt[3]{3}}{2} \cdot \frac{6x^{\frac{5}{6}}}{5} + c = \frac{\sqrt{2}x^2}{4} - \frac{3\sqrt{2}\sqrt[3]{3}\sqrt[6]{x^5}}{5} + c$$

References